

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

150 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

150 KWP GRID-TIED SOLAR PLANT

LT CATEGORY II — COMMERCIAL

Net investment ₹61.43 L after input tax credit, generating 2,29,950 units in year one against an energy charge of ₹10.15 per unit.

30-YEAR ANALYSIS

PAYBACK

3.0 yrs

Post-tax, undiscounted

PROJECT IRR

33.2%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.32

per kWh, against ₹10.15 from the grid

NET PRESENT VALUE

₹1.54 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	150 kWp
Year 1 generation	2,29,950 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	LT Category II — Commercial (LT supply)
Energy charge offset	₹10.15 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹21.64 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹43.00 / Wp (incl. GST)
Project price incl. GST	₹64.50 L
GST at 5% — recovered as ITC	₹3.07 L
Net capitalised cost	₹61.43 L
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹15.46 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	61,42,857	24,57,143	36,85,714	6,18,414
2	36,85,715	14,74,286	22,11,429	3,71,048
3	22,11,428	8,84,571	13,26,857	2,22,629
4	13,26,857	5,30,743	7,96,114	1,33,577
5	7,96,115	3,18,446	4,77,669	80,146
6	4,77,668	1,91,067	2,86,601	48,088
7	2,86,601	1,14,640	1,71,961	28,853
8	1,71,960	68,784	1,03,176	17,312
9	1,03,177	41,271	61,906	10,387
10	61,906	24,762	37,144	6,232
11	37,143	14,857	22,286	3,739
12	22,286	8,914	13,372	2,243
Tax shield, years 1-12				15,42,669

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30-year post-tax cash flow

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How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	2,29,950	21,63,830	—	24,57,143	(73,821)	22,37,651	22,37,651
2	2,28,800	22,39,131	90,000	14,74,286	1,69,845	19,79,286	42,16,937
3	2,27,656	23,17,053	90,000	8,84,571	3,37,876	18,89,177	61,06,114
4	2,26,518	23,97,686	90,000	5,30,743	4,47,221	18,60,465	79,66,579
5	2,25,385	24,81,125	90,000	3,18,446	5,21,652	18,69,473	98,36,052
6	2,24,258	25,67,469	90,000	1,91,067	5,75,441	19,02,027	1,17,38,079
7	2,23,137	26,56,816	90,000	1,14,640	6,17,164	19,49,653	1,36,87,732
8	2,22,021	27,49,274	90,000	68,784	6,51,974	20,07,299	1,56,95,031
9	2,20,911	28,44,948	90,000	41,271	6,82,978	20,71,970	1,77,67,001
10	2,19,807	29,43,953	90,000	24,762	7,12,051	21,41,902	1,99,08,903
11	2,18,708	30,46,402	90,000	14,857	7,40,328	22,16,074	2,21,24,977
12	2,17,614	31,52,417	90,000	8,914	7,68,506	22,93,911	2,44,18,888
13	2,16,526	32,62,121	90,000	5,349	7,97,013	23,75,108	2,67,93,996
14	2,15,444	33,75,643	90,000	3,209	8,26,123	24,59,520	2,92,53,516
15	2,14,366	34,93,115	90,000	1,926	8,56,011	25,47,104	3,18,00,620
16	2,13,294	36,14,676	90,000	1,155	8,86,800	26,37,876	3,44,38,496
17	2,12,228	37,40,466	90,000	693	9,18,575	27,31,891	3,71,70,387
18	2,11,167	38,70,635	90,000	416	9,51,405	28,29,229	3,99,99,616
19	2,10,111	40,05,333	90,000	250	9,85,348	29,29,985	4,29,29,601
20	2,09,060	41,44,718	90,000	150	10,20,454	30,34,264	4,59,63,865
21	2,08,015	42,88,954	90,000	90	10,56,770	31,42,184	4,91,06,049
22	2,06,975	44,38,210	90,000	54	10,94,344	32,53,866	5,23,59,915
23	2,05,940	45,92,660	90,000	32	11,33,221	33,69,439	5,57,29,354
24	2,04,911	47,52,484	90,000	19	11,73,449	34,89,035	5,92,18,389
25	2,03,886	49,17,871	90,000	12	12,15,076	36,12,795	6,28,31,184
26	2,02,867	50,89,013	90,000	7	12,58,150	37,40,863	6,65,72,047
27	2,01,852	52,66,110	90,000	4	13,02,722	38,73,388	7,04,45,435
28	2,00,843	54,49,371	90,000	3	13,48,846	40,10,525	7,44,55,960
29	1,99,839	56,39,009	90,000	2	13,96,574	41,52,435	7,86,08,395
30	1,98,840	58,35,247	90,000	1	14,45,963	42,99,283	8,29,07,678
Total	64,20,929	11,13,35,740	26,10,000	61,42,856	2,58,18,059	8,29,07,678	8,29,07,678

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

150 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.0 yrs
Discounted payback	3.8 yrs
Project IRR, unlevered	33.2%
Net present value at 10%	₹1.54 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.32 / kWh
Cumulative return, 30 years	1250%
Total net cash flow, after tax	₹8.29 Cr
30-year generation	64,20,931 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.32
Grid energy charge today		₹10.15

Every unit the plant generates costs **₹3.32** across its life against **₹10.15** bought from APSPDCL today — and the grid figure reaches **₹31.65** by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 64,20,931 units, avoiding 5,265.2 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** LT Category II — Commercial at ₹10.15/kWh. Marginal rate, above 500 units per month. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹64.50 L.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 150 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.